

MMT-04 Summary Document

Overview

This document summarizes the lecture MMT-04 on Algorithmic Trading Process.

The lecture covers the following topics:

- Investment Management
- Trading Costs
- Market Impact Model and its Development
- Developing a Market Impact Model
- Trading Analytics - Portfolio Pre-Trade Model
- Cost Curves

Investment Management

There are 2 styles:

1. Active: which, in turn, can be:
 - a. Fundamental: Uses financial statements and corporate annual reports
 - b. Quantitative: Uses backtesting and portfolio optimization
2. Passive: It only follows an index.

Investors' investment motivations can have multiple reasons. The following are some of the most important and are not comprehensive:

- Short-term alpha: When you have a short-term view (e.g. minutes or hours) of the market and you invest accordingly.
- Long-term alpha: When you have a view higher than a month of the market and you invest based on that horizon.
- Stock mispricing: When you have a view of the fundamental value of a stock.
- Portfolio optimization: Stock selection and position size are based on optimization methods that follow certain risk/return targets.
- Portfolio rebalance: In order to attain a certain risk/reward target, you - also rebalance the stock allocation.
- Index change: When you trade based on index changes. Depending on the changes (index rebalancing, merge and acquisition changes, company bankruptcy, etc.), you make changes in your trading.
- Cash inflow: When you receive an inflow of money from your investors, you then proceed to invest it.
- Cash redemption: Investors' consumption needs are important to be taken into account while setting the investment objectives.
- Income: Dividends that are paid from holding stocks and you decide to invest in the same stocks or other stocks or asset classes.

Trader's Dilemma

The more aggressive the trading, the more impact the trader will have on the market and the lower the timing risk.
The more passive the trading, the higher the price risk the trader will face and the lower the market impact cost.

The trader should find an optimum between her market impact cost (MI) and the timing risk (TR). This can be accomplished with a loss function.

Mathematically speaking, we can create this loss function as

$$\text{Min} (MI + \lambda * TR)$$

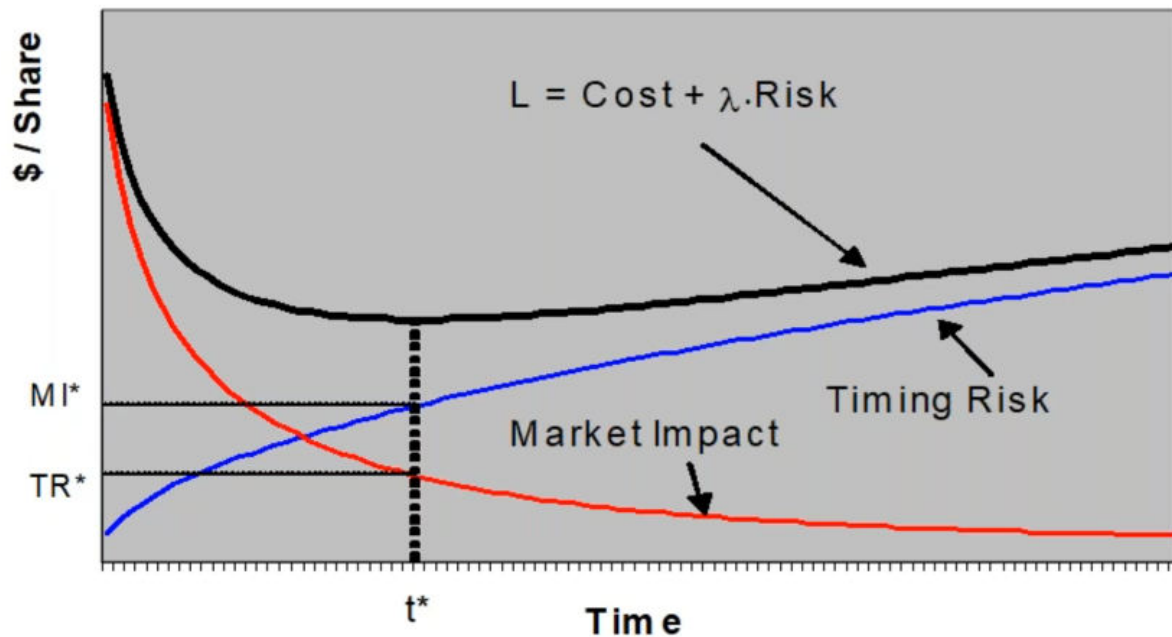
Where λ is the trader's risk aversion. We need to minimize the function to find an optimum.

The market impact can be understood as the change in price caused by trade or order. This cost is often the main reason managers lag behind their peers. Market impact costs can occur for two reasons:

- Liquidity needs and urgency demands (Temporary cause)
- Information content: Price information that market participants use to adjust their prices to this new information fair value.

The more risk-averse the trader, the more aggressive the trader will trade. The less risk-averse, the more passively trader will trade.

Single Stock Optimization



Trading Cost Measurement

A model to compute the trading costs is the Implementation Shortfall: This can be understood as: Costs, fees, commissions, and opportunity costs. Some other costs are:

- Arrival cost: The price value you saved in your trading app before you trade.
- Trade cost: The price you agreed to pay.

- Execution cost: The price you achieved to pay for trading the asset.

These 3 prices not necessarily are equal because prices change in markets within a fraction of a second.

Trade Performance

Trade performance done by your broker to give the best prices can be evaluated by:

1. Benchmark Analysis: Measures the simplest of the trading cost analysis performance evaluation techniques. These are intended to compare specific measures such as net difference and tracking error or to distinguish between temporary and permanent impact.
 - a. Pre-Trade Performance: Provides a measure of trading cost. Commonly used are the following:
 - i. Arrival Price: Stock price at the time the order entered the market.
 - ii. Open Price: as a proxy for arrival price.
 - b. Intra-Day Performance: Provides a measure of trading performance.
 - i. VWAP: The volume weighted average price over the day.
 - ii. Interval VWAP: The VWAP over the trading horizon.
 - c. Post-Trade Performance: Provides a measure of trading profitability.
 - i. Close: Useful to compute the end-of-day tracking error and is more commonly used by index funds that use the closing price in the valuation of the fund.
 - ii. Next Day's Open
 - iii. Next Day Close or Future Day Close
2. VWAP
3. RPM: Relative performance measure
4. Value-Add
5. Z-Score

In order to forecast the trading cost, you can:

- Estimate the market impact costs and timing risk.
- Build a trading cost distribution.

Implementation Shortfall

Before we get into this, you should understand the following:

- Delay Related: Price movement in the stock from the time of the investment decision until the order is released to the market.
- Trading Related: Price movement in the stock over the time the order is being executed in the market.
- Opportunity Cost: The missed profit opportunity of not being able to execute all shares in the order.

An equation to compute implementation shortfall is the following:

$$Expanded\ IS = \underbrace{S(P_0 - P_d)}_{Delay\ Related} + \underbrace{\left(\sum s_j\right)(P_{avg} - P_0)}_{Trading\ Related} + \underbrace{\left(S - \sum s_j\right)(P_n - P_0)}_{Opportunity\ Cost} + Fixed$$

Where:

S = Order Shares

$\sum s_j$ = Shares Traded

$S - \sum s_j$ = Unexecuted Shares

P_{avg} = Average execution price of the order (for all traded shares).

P_0 = Arrival Price (mid-point of bid-ask spread at the time the order was entered into the market)

P_d = Decision Price (mid-point of bid-ask spread at the time of the investment decision)

P_n = End Price (mid-point of bid-ask spread at end of trading or closing price on the day. The end of trading could occur at some time in the future, e.g., day 2 or day 3 or later.)

Fixed = Commissions, Taxes, and Fees & Rebates

Spread Costs are embedded in the actual execution price of the stock

This equation can help you understand which broker gives the best immediate price value when you trade.

Benchmark analysis

It is done using the following formula:-

$$\text{Benchmark Cost} = \text{Side} \cdot \frac{P_{avg} - P_B}{P_B} \cdot 10^4_{bp}$$

Where,

P_{avg} = Average Execution Price

P_B = Price Benchmark

$\text{Side} = 1$ if Buy, -1 if Sell

Relative Performance Measure

It's a percentile ranking of trading activity. It will provide you with an indication of the percentage of total activity that the investor outperformed in the market. For the buy (sell) order, it represents the percentage of market activity that is transacted at a higher (lower) price.

A descriptive statistic formula used in standardised academic tests is the following:

$$RPM_{Buy} = \frac{1}{2} \cdot \left(\frac{\text{Total Volume} + \text{Volume at Price} > P_{Avg} - \text{Volume at Price} < P_{Avg}}{\text{Total Volume}} \right)$$

$$PM_{Sell} = \frac{1}{2} \cdot \left(\frac{\text{Total Volume} + \text{Volume at Price} < P_{Avg} - \text{Volume at Price} > P_{Avg}}{\text{Total Volume}} \right)$$

The Z-Score

Provides a risk-adjusted performance score by normalizing the difference between estimated and actual values by the timing risk of the execution.

A commonly used measure is the following:

$$Z = \frac{Actual - Expected}{Standard Deviation}$$

$$Z = \frac{PreTrade Cost Estimate - Arrival Cost}{PreTrade Timing Risk}$$

$Z \sim (0,1)$

Value-Add

Used to determine if the realized transaction costs were appropriately given the actual market conditions during the trading horizon. Calculated based on the arrival price benchmark cost and the expected market impact cost, where the latter is based on the actual market conditions.

The commonly used formula is the following:

$$ValueAdd = Est. Market Impact - Arrival Cost$$

A positive (negative) ValueAdd indicates out-(under-)performance.

Market Impact Model

Multiple factors can explain the market impact cost. Let's see some of them:

- Order size: Size as a percentage of the average daily volume.
- Volatility: price sensitivity to traded shares variation in the stock market.
- POV: Percentage of Volume or rate at which the asset trades in the market.
- Market capitalization: Large (small) capitalization stocks have lower (higher) impact costs.

The I-Star Model

This model serves as the main model to analyze market impact cost.

The model formulation is comprised of 3 equations and is presented below:

$$I_{bp}^* = a_1 * Size^{a_2} * \sigma^{a_3}$$
$$MI_{bp} = b_1 I^* * POV^{a_4} + (1 - b_1) * I^*$$

$$TR = \sigma * \sqrt{\frac{1}{250} * \frac{1}{3} * \frac{S}{ADV} * \frac{1 - POV}{POV}} * 10^4_{bp}$$

Where:

I_{bp}^* : Instantaneous market impact cost and represents the cost an

investor would incur if they released the entire order to the market for execution at one time.

MI_{bp} : Market impact cost of the order and is dependent upon the execution strategy, e.g., POV rate used to transact the shares. This formulation of market impact cost also distinguishes between temporary impact due to the liquidity needs of the fund and permanent market impact due to the information content of the trade.

Size: Order size expressed as shares to trade divided by the stock's average daily volume, expressed as a decimal.

σ : Annualized price volatility, expressed as a decimal, i.e., 20% is expressed as 0.20.

POV: Volume participation rate, i.e., shares traded divided by the volume in the trading period, expressed as a decimal.

a_1 : sensitivity to trade size, i.e., scaling factor.

a_2 : order shape parameter.

a_3 : volatility shape parameter.

a_4 : percentage of the volume shape parameter, allows for a higher level of flexibility in an ever-changing market.

b_1 : percentage of temporary market impact, i.e., an indication of the liquidity cost component cost, with $0 \leq b_1 \leq 1$.

$(1 - b_1)$: percentage of permanent market impact, i.e., an indication of the cost due to the information content of the trade, $0 \leq (1 - b_1) \leq 1$.

TR: Timing risk.

ADV: Average daily volume.

S: Number of shares in the order.

This model can be simplified by assuming the permanent and temporary impact terms are zero. Thus, the model is reduced to

$$I_{bp}^* = a_1 * Size^{a_2} * \sigma^{a_3} * POV^{a_4}$$

The timing risk value can be obtained without any estimation. You just need to estimate the first two equations' parameters.

There are 3 approaches to estimating the market impact parameters:

- Log-Linear Model
- Iterative Solution
- Non-Linear Model